

SPEECH EMOTION CLASSIFICATION

Comprehensive Exploratory Data Analysis & Statistical Quality Report

Project: Group 02 (G2) ML Lab Mini-Project | **Client/Domain:** CallConnect Speech / Telephony Customer Support

Dataset: RAVDESS Speech 16 kHz | **Stage:** Data Analysis (DA) Technical Audit

1. Executive Objective & Operational Framing

In automated customer-support telephony environments (CallConnect Speech), inbound call routing and real-time supervisor escalation depend fundamentally on reliable **Speech Emotion Classification (SEC)**. Classifying acoustic vocalizations into affective states (such as distress, anger, frustration, satisfaction, and neutral baseline) enables proactive churn mitigation and agent support. Following the initial Data Engineering stage executed by Kenisha, this Data Analysis technical audit rigorously interrogates the processed dataset (dataset_metadata.csv) to validate acoustic distributions, audit speaker balance, assess audio signal properties, detect anomalies, formulate statistical hypotheses, and establish **three high-priority actionable insights** vital for downstream Data Science modeling.

2. Dataset Architectural Overview & Quality Verification

The dataset consists of the official 16 kHz speech subset of the **Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)**. A total of **1,440 WAV speech recordings** were ingested and parsed into structured metadata across 16 schema columns. Every file represents one of two lexically neutral statements ('Kids are talking by the door', 'Dogs are sitting by the door') vocalized across 8 categorical emotional states by 24 professional North American actors.

Metric / Dimension	Empirical Value	Verification Status / Impact
Total Audio Recordings	1,440 files	100% complete; 0 missing or corrupted files
Professional Actors	24 actors (12 Female, 12 Male)	Exact 50/50 gender balance across dataset
Utterances per Actor	60 recordings / actor	Standard deviation = 0.0 (perfectly uniform)
Target Emotional States	8 classes (Neutral, Calm, Happy, etc.)	Categorical classification objective
Digital Sampling Rate	16,000 Hz (16 kHz)	100% uniform; zero resampling required
Vocal Channel Architecture	1 Channel (Mono)	100% uniform; zero channel downmixing needed
Signal Duration Range	2.9363 s – 5.2719 s	Mean: 3.7007 s, Median: 3.6704 s, Std: 0.3367 s
Train / Test Split Ratio	1,140 train (79.2%) / 300 test (20.8%)	Actor-grouped split; 0 speaker overlap
Data Quality & Missing Values	0 missing, 0 duplicates	Flawless structural data completeness

3. Emotion Class Distribution Analysis

A rigorous analysis of sample frequencies across all 8 target emotion classes reveals an essential structural nuance: the dataset is **approximately balanced** across 7 emotions, but features an intentional **1:2 under-representation of neutral speech**.

Emotion Code	Emotion Name	Sample Count	Percentage	Intensity Levels Represented
1	neutral	96	6.67%	Normal Only (Intensity = 1)

2	calm	192	13.33%	Normal (96) & Strong (96)
3	happy	192	13.33%	Normal (96) & Strong (96)
4	sad	192	13.33%	Normal (96) & Strong (96)
5	angry	192	13.33%	Normal (96) & Strong (96)
6	fearful	192	13.33%	Normal (96) & Strong (96)
7	disgust	192	13.33%	Normal (96) & Strong (96)
8	surprised	192	13.33%	Normal (96) & Strong (96)
TOTAL	8 Emotions	1,440	100.00%	All Standard Trials

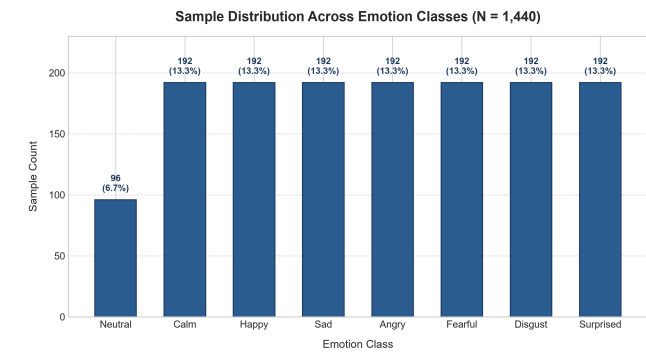


Figure 1: Sample count by emotion class.

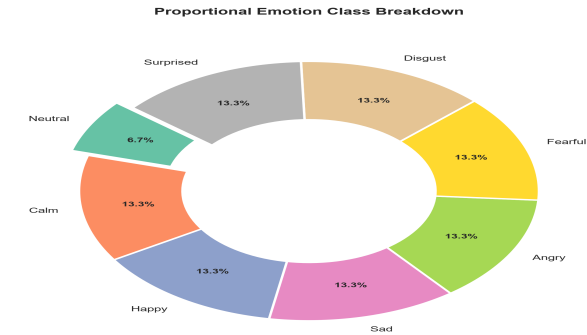


Figure 2: Proportional class breakdown.

4. Audio Signal Characteristics & Duration Outlier Analysis

Utterance duration dictates temporal framing, feature matrix dimensionality, padding limits, and temporal pooling strategies. The overall dataset duration spans **2.9363 seconds to 5.2719 seconds**, with an empirical mean of **3.7007 s** and median of **3.6704 s** (standard deviation = 0.3367 s).

Statistical Parameter	Value (Seconds)	Value (Sample Count at 16 kHz)
Minimum Duration	2.9363 s	46,981 samples
First Quartile (Q1 - 25%)	3.4702 s	55,523 samples
Median (Q2 - 50%)	3.6704 s	58,727 samples
Mean (Parametric Average)	3.7007 s	59,211 samples
Third Quartile (Q3 - 75%)	3.8706 s	61,930 samples
Maximum Duration	5.2719 s	84,350 samples
Interquartile Range (IQR)	0.4004 s	6,407 samples
Tukey Upper Bound (Q3 + 1.5×IQR)	4.4712 s	71,539 samples

Statistical Outlier Investigation (N = 37):

Applying Tukey's conservative rule ($Q3 + 1.5 \times IQR$) identifies exactly **37 audio files (\$2.571\%**) with duration $> 4.4712\text{ s}$. A deep forensic examination reveals that **35 of the 37 outliers (\$94.61\%)** are from **Intensity 2 (strong emotional intensity)**, with 17 belonging to *disgust*, 10 to *calm*, and 5 to *sad*. Actor 19 contributed 8 of these utterances. Crucially, these recordings are **acoustically pristine, non-corrupted human speech** representing prolonged emotional articulation. Deleting them would systematically strip out high-intensity emotional expressions. Under Kenisha's fixed operational threshold ($2.0\text{ s} - 6.0\text{ s}$), zero files were flagged.

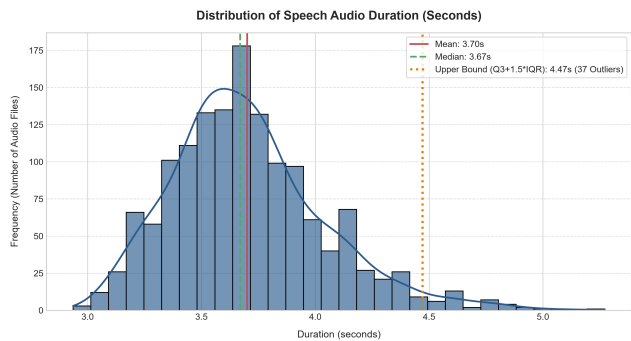


Figure 3: Duration distribution histogram with KDE & IQR bounds.

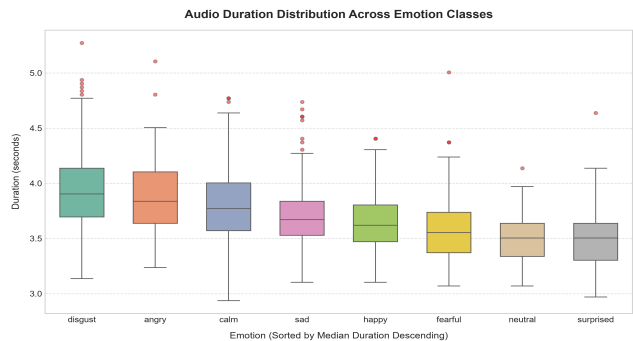


Figure 4: Duration distribution by emotion class.

5. Actor Representation & Speaker Leakage Audit

In speech analytics, **speaker identity** is the single greatest source of data leakage. When recordings from the same speaker appear in both training and test sets, machine learning models exploit speaker-specific pitch (F_0), vocal tract resonances, and idiosyncratic timbre, producing artificially inflated validation accuracy that collapses during real-world inference. Kenisha's Data Engineering pipeline enforced a strict **Actor-Grouped Split** (`GroupShuffleSplit`) with 19 actors in train and 5 in test, ensuring **zero actor overlap**. Furthermore, every actor contributes exactly 60 recordings with an identical emotion breakdown (4 neutral, 8 of every other emotion).

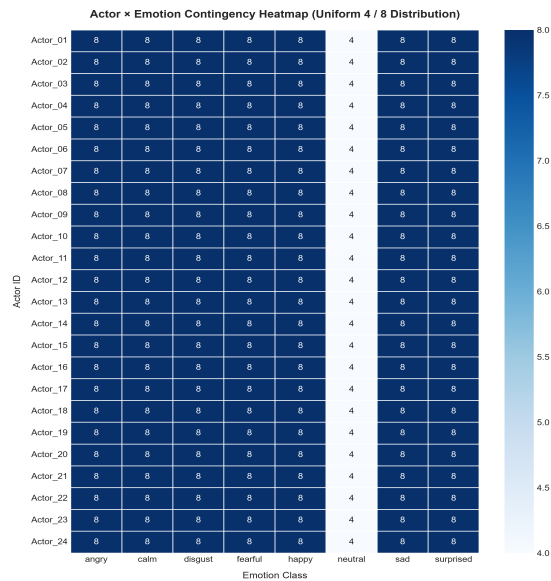


Figure 5: Actor x Emotion contingency heatmap (Uniform 4/8 cells).

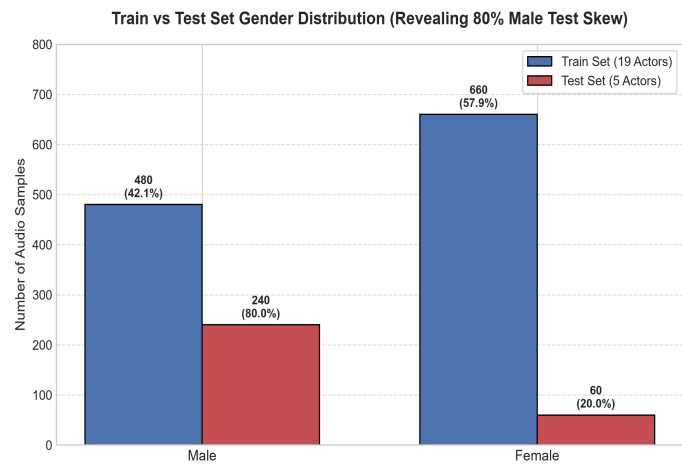


Figure 6: Train vs Test split gender distribution (80% Male test skew).

6. Inferential Statistical Analysis & Hypothesis Testing

To establish rigorous empirical relationships between acoustic features and experimental conditions, we conducted inferential hypothesis tests:

- 1. One-Way ANOVA across Emotion Classes:** Testing whether audio duration differs across emotions yielded $F = 55.47$, $p = 2.21 \times 10^{-70}$. This confirms highly significant duration variation: *disgust* (3.9418s) and *angry* (3.8714s) are systematically longer, whereas *surprised* (3.4875s) and *neutral* (3.5032s) are shortest.
- 2. Two-Sample T-Test across Intensity Tiers:** Utterances delivered at strong intensity (Intensity 2) have an empirical mean duration of 3.8238s, compared to 3.5929s for normal intensity ($t = -13.81$, $p = 7.59 \times 10^{-41}$). Strong emotional arousal elongates speech articulation by an average of **231 milliseconds**.
- 3. Correlation Analysis:** Duration and discrete samples exhibit a deterministic linear correlation ($r = 1.0000$). Intensity displays a moderate positive correlation with duration ($r = 0.3423$, $p < 0.001$), while statement sentence and repetition show near-zero correlation ($|r| < 0.07$).

7. Comprehensive Anomaly Audit

A systematic scan was executed across all 1,440 audio records to identify corrupted signals, out-of-range metadata, and schema violations:

- Signal Integrity:** 0 unreadable audio files, 0 corrupted headers, 0 infinite/NaN feature values.
- Sampling & Channels:** 100% adherence to 16,000 Hz sample rate; 100% adherence to single-channel Mono.
- Metadata & Uniqueness:** 0 missing fields across all 16 attributes; 0 duplicate filenames or file paths.
- Subtle Anomaly Detected (Gender Skew in Split):** The unstratified group split allocated 4 male actors and 1 female actor to the test set, creating an **80% male test set** that skews downstream evaluation.

8. Three Prescriptive Actionable Insights for Machine Learning

INSIGHT 1: Neutral Emotion Class Under-Representation (1:2 Imbalance)

Finding: The dataset exhibits an inherent 1:2 class under-representation for neutral relative to all other 7 emotion classes.

Evidence: neutral contains exactly 96 samples (6.67%), whereas each other emotion contains 192 samples (13.33%). This occurs because RAVDESS excludes strong intensity for neutral speech.

Why it Matters: In CallConnect telephony workflows, neutral speech is the dominant conversational baseline. An unweighted classifier will suffer lower prior probability and poor recall on neutral speech.

Prescriptive Action: Implement class-weighted cross-entropy loss ($w_{\text{neutral}} = 2.0$, $w_{\text{others}} = 1.0$) and adopt Macro-F1 as the primary model selection criterion.

INSIGHT 2: Severe Gender Skew in Actor-Grouped Train/Test Split (80% Male Test Split)

Finding: The actor-grouped split created zero speaker leakage but produced an acute gender skew between train and test sets.

Evidence: Test set contains 240 male samples (80.0%) across 4 actors (1, 9, 17, 19) and only 60 female samples (20.0%) from Actor 12. Train set is female-dominated (660 female, 57.9% vs 480 male, 42.1%).

Why it Matters: Vocal tract geometry creates distinct fundamental frequency (F_0) and formant baselines between genders. An 80% male test set yields performance metrics biased toward male speakers and masks female generalization failure.

Prescriptive Action: The Data Scientist must implement **Gender-Stratified Group Splitting** (e.g., allocating 2 female and 2-3 male actors to test, or running 6-fold cross-validation stratified by gender) to enforce a 50/50 gender ratio.

INSIGHT 3: Emotional Intensity Drives Duration & Explains Prosodic Outliers

Finding: Utterance duration is governed by emotional intensity and emotion type, generating 37 statistical outliers that represent genuine expressive elongation rather than corrupted data.

Evidence: High intensity utterances are on average 231 ms longer than normal intensity ($p = 7.59 \times 10^{-41}$). 94.6% of statistical duration outliers occur in strong intensity recordings, heavily concentrated in *disgust* and *calm*.

Why it Matters: Fixed-frame audio architectures require uniform tensor dimensions. Truncating signals to median duration (3.67 s) or deleting outliers clips and discards the most potent affective cues.

Prescriptive Action: Retain all 37 outlier files. Standardize 2D spectrogram tensor shapes using right-side zero-padding to 5.30 s (84,800 samples), or use time-pooled statistical feature summaries (mean, std of MFCCs over time).

9. Prescriptive Recommendations for the Data Scientist

- Gender-Stratified Split:** Reconfigure the test partition to guarantee 2 female and 2 male actors (or 3 male / 3 female 4-fold cross validation) to ensure unbiased gender evaluation.
- Input Padding Envelope:** Pad all raw signals to 5.30 seconds (84,800 samples at 16 kHz) with trailing zeros for spectrogram generation, avoiding truncation of the 37 long-duration emotional outliers.
- Feature Representation:** When extracting tabular features, concatenate mean and standard deviation across time frames for 40 MFCCs, Delta-MFCCs, and Spectral Contrast (producing a compact 128-dimensional vector).
- Loss Function Balancing:** Utilize weighted cross-entropy with a 2.0x penalty factor assigned to neutral speech to compensate for its half-representation.
- Evaluation Protocol:** Report Macro-Averaged F1-Score, Class-Wise Recall, and an Actor-Stratified Confusion Matrix as formal project success criteria.

10. Conclusion & Handover Protocol

The Data Analysis phase has conclusively verified that the RAVDESS 16 kHz dataset is of exemplary acoustic quality, containing zero corrupted recordings and perfect 16 kHz mono consistency. With the structural half-representation of neutral speech identified, the acoustic duration dynamics quantified, and the gender skew in the test split exposed, the modeling foundation is fully characterized and ready for seamless handover to the Data Science stage.